35—BAR SCALES (continued)

BAR SCALE CALCULATIONS — FEET (1 foot = 12 inches)

FRACTIONAL SCALE	SCALE TO MAP REPRESENTATION: Scale Unit :	represents Map Unit	TO FIND FEET PER INCH (**x** in ratio) Use ratio below or $\frac{SCALE}{12}$	FEET PER INCH	TOTAL FEET ON SCALE	TO FIND TOTAL SCALE LENGTH IN INCHES (**y** in ratio) Use ratio below or $\frac{\text{Feet on scale}}{\text{Feet per inch}}$	TOTAL SCALE LENGTH (INCHES)
1:12 000	1 inch	12 000 in	$\frac{12}{1} = \frac{12\,000}{\mathbf{x}}$	1000.000	6000	$\frac{1000.000}{1} = \frac{6000}{\mathbf{y}}$	6.000
1:20 000	1 inch	20 000 in	$\frac{12}{1} = \frac{20\,000}{\mathbf{x}}$	1666.6666	8000	$\frac{1666.6666}{1} = \frac{8000}{\mathbf{y}}$	4.800
1:24 000	1 inch	24 000 in	$\frac{12}{1} = \frac{24\,000}{\mathbf{x}}$	2000.000	8000	$\frac{2000.000}{1} = \frac{8000}{\mathbf{y}}$	4.000
1:25 000	1 inch	25 000 in	$\frac{12}{1} = \frac{25\,000}{\mathbf{x}}$	2083.3333	8000	$\frac{2083.3333}{1} = \frac{8000}{\mathbf{y}}$	3.840
1:31 250	1 inch	31 250 in	$\frac{12}{1} = \frac{31\,250}{\mathbf{x}}$	2604.1666	8000	$\frac{2604.1666}{1} = \frac{8000}{\mathbf{y}}$	3.072
1:31 680	1 inch	31 680 in	$\frac{12}{1} = \frac{31\,680}{\mathbf{x}}$	2640.000	8000	$\frac{2640.000}{1} = \frac{8000}{\mathbf{y}}$	3.030
1:48 000	1 inch	48 000 in	$\frac{12}{1} = \frac{48\,000}{\mathbf{x}}$	4000.000	24 000	$\frac{4000.000}{1} = \frac{24\,000}{\mathbf{y}}$	6.000
1:50 000	1 inch	50 000 in	$\frac{12}{1} = \frac{50\,000}{\mathbf{x}}$	4166.6666	24 000	$\frac{4166.6666}{1} = \frac{24\,000}{\mathbf{y}}$	5.760
1:62 500	1 inch	62 500 in	$\frac{12}{1} = \frac{62\,500}{\mathbf{x}}$	5208.3333	30 000	$\frac{5208.3333}{1} = \frac{30\,000}{\mathbf{y}}$	5.760
1:63 360	1 inch	63 360 in	$\frac{12}{1} = \frac{63\,360}{\mathbf{x}}$	5280.000	30 000	$\frac{5280.000}{1} = \frac{30\,000}{\mathbf{y}}$	5.681
1:72 000	1 inch	72 000 in	$\frac{12}{1} = \frac{72\,000}{\mathbf{x}}$	6000.000	30 000	$\frac{6000.000}{1} = \frac{30\,000}{\mathbf{y}}$	5.000
1:75 000	1 inch	75 000 in	$\frac{12}{1} = \frac{75\,000}{\mathbf{x}}$	6250.000	30 000	$\frac{6250.000}{1} = \frac{30\,000}{\mathbf{y}}$	4.800
1:96 000	1 inch	96 000 in	$\frac{12}{1} = \frac{96\,000}{\mathbf{x}}$	8000.000	33 000	$\frac{8000.000}{1} = \frac{33\,000}{\mathbf{y}}$	4.125

To find feet per inch on 1: 12 000 map . . .

12 inches = 1 foot
Show in ratio as ...

$$\frac{12}{1}\ \frac{\textit{inches}}{\textit{feet}}$$

Let SCALE (12 000) be in inches
Fractional scale says 1 inch represents 12,000 in
Let **x** be feet that 1 inch represents on map
Show in ratio as ...

$$\frac{12\,000}{\mathbf{x}}\ \frac{\textit{inches}}{\textit{feet}}$$

Solution . . .

$$\frac{12}{1} = \frac{12\,000}{\mathbf{x}}$$

$$12 \cdot \mathbf{x} = 12\,000 \cdot 1$$

$$\frac{12\,\mathbf{x}}{12} = \frac{12\,000}{12}$$

$$\mathbf{x} = \frac{12\,000}{12}\ \text{(SCALE)}$$

$$\mathbf{x} = 1000.00$$